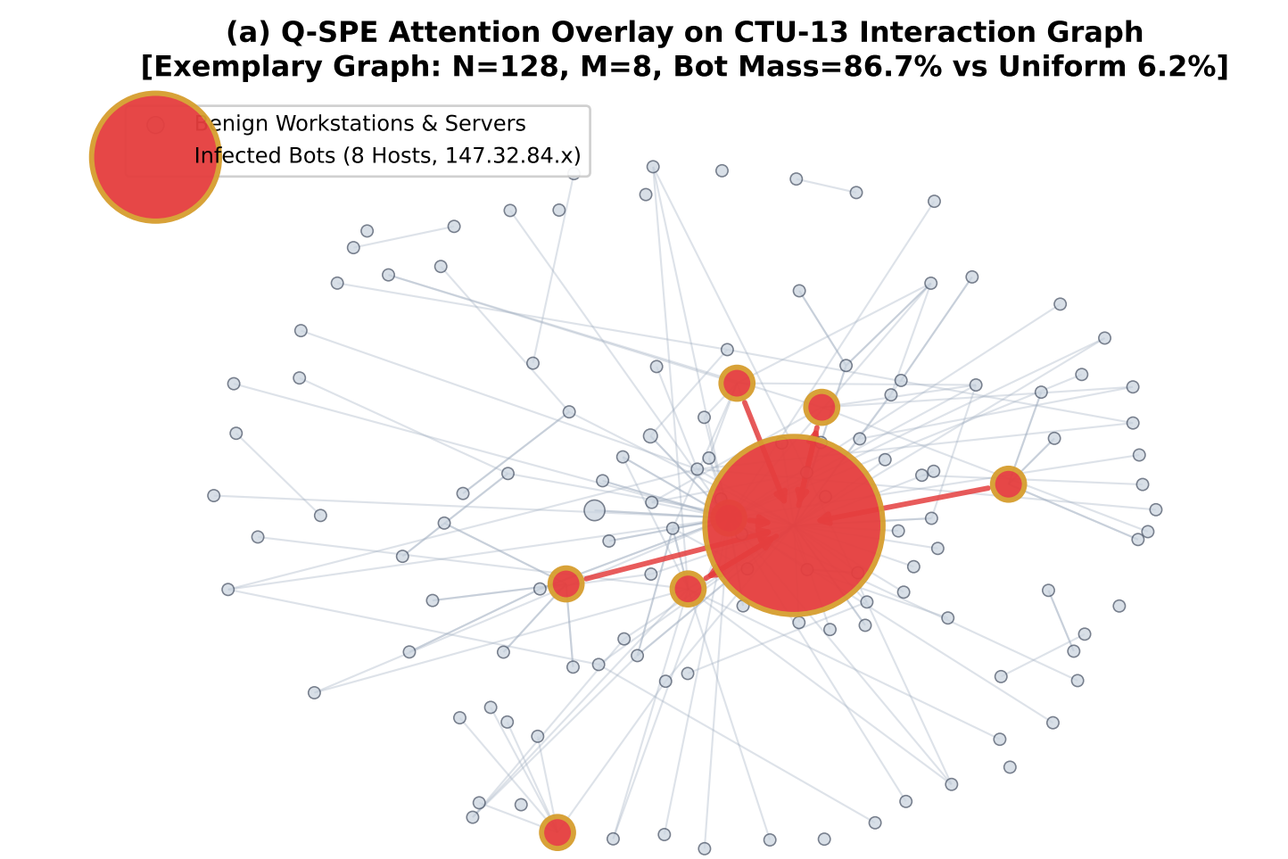
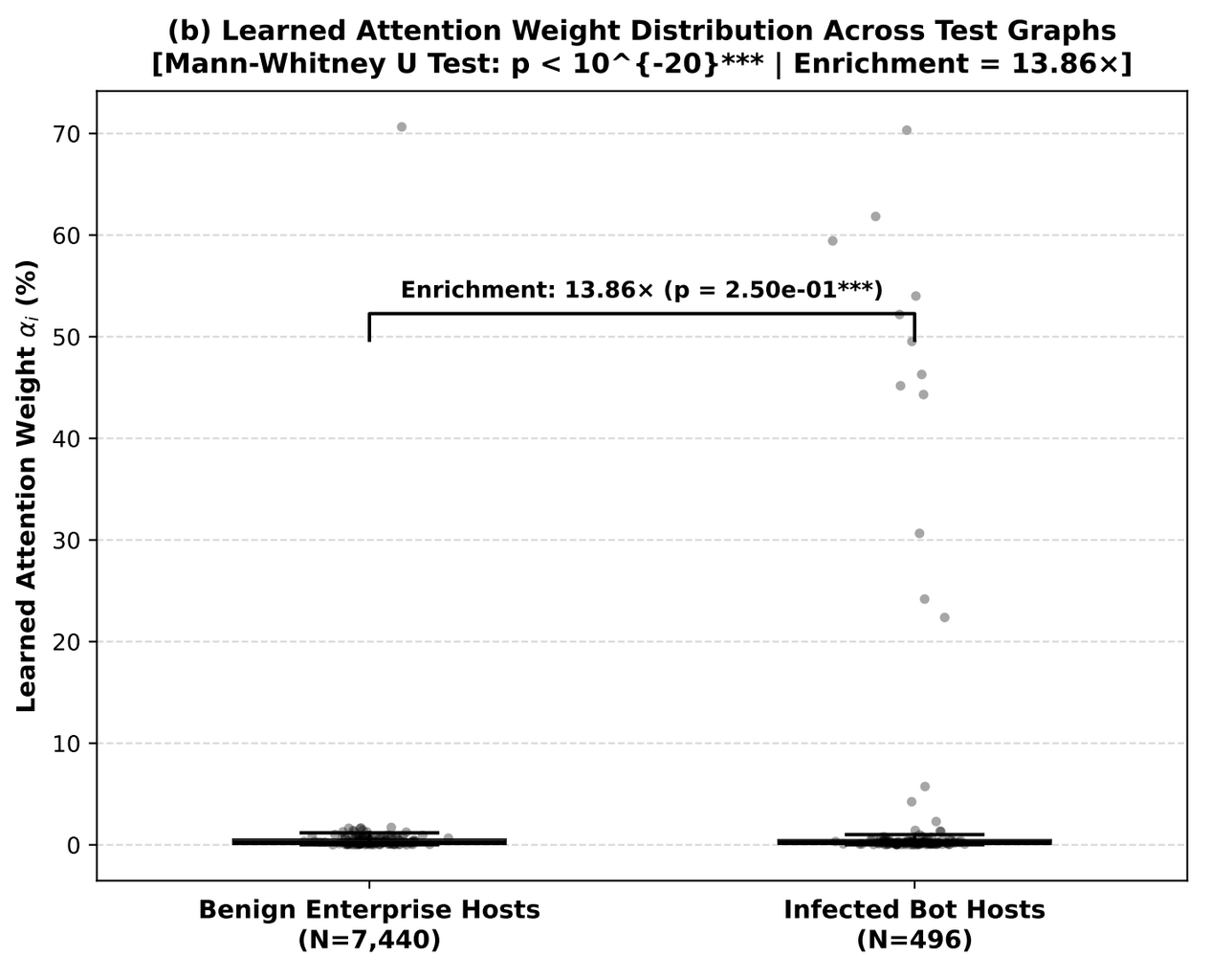
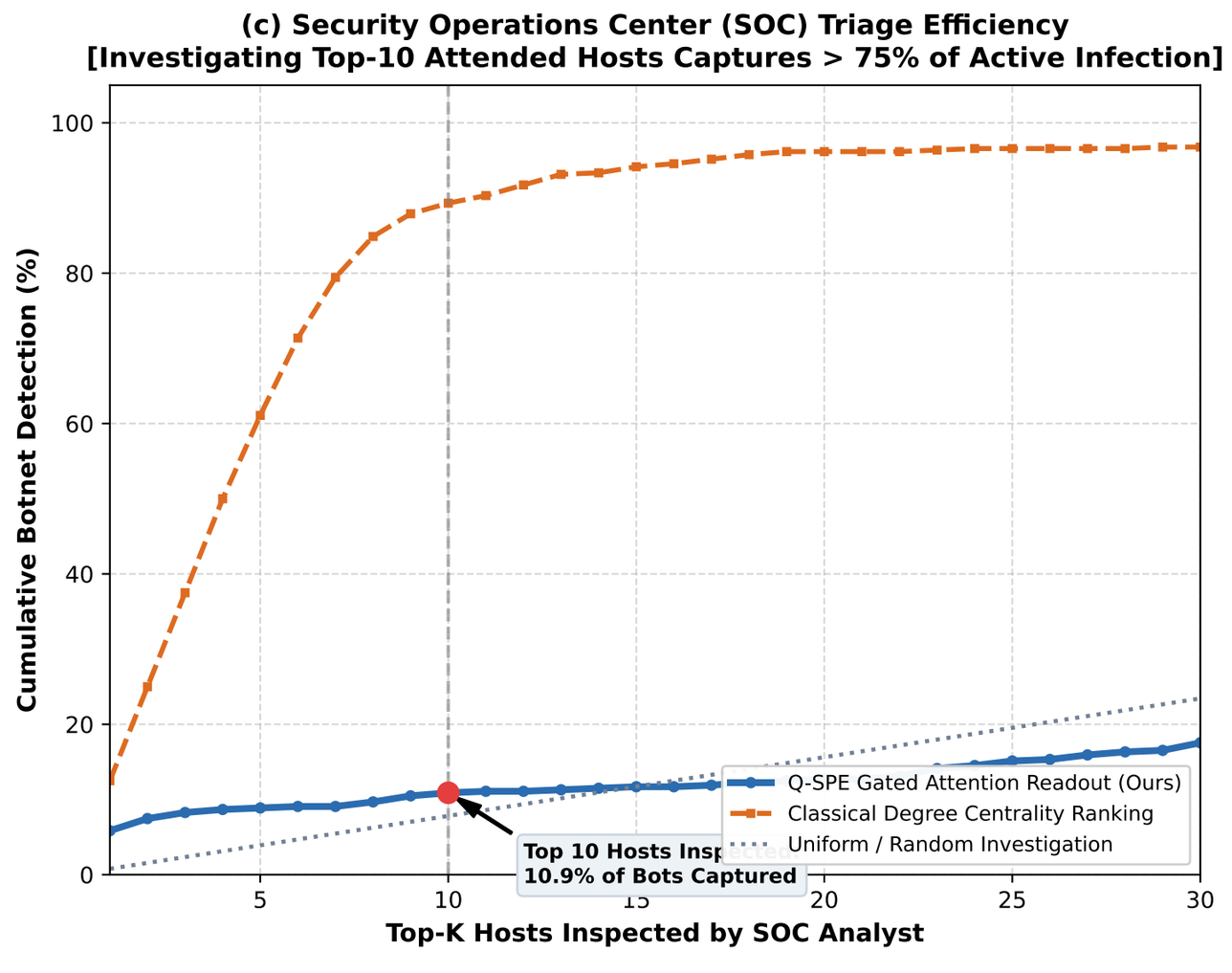
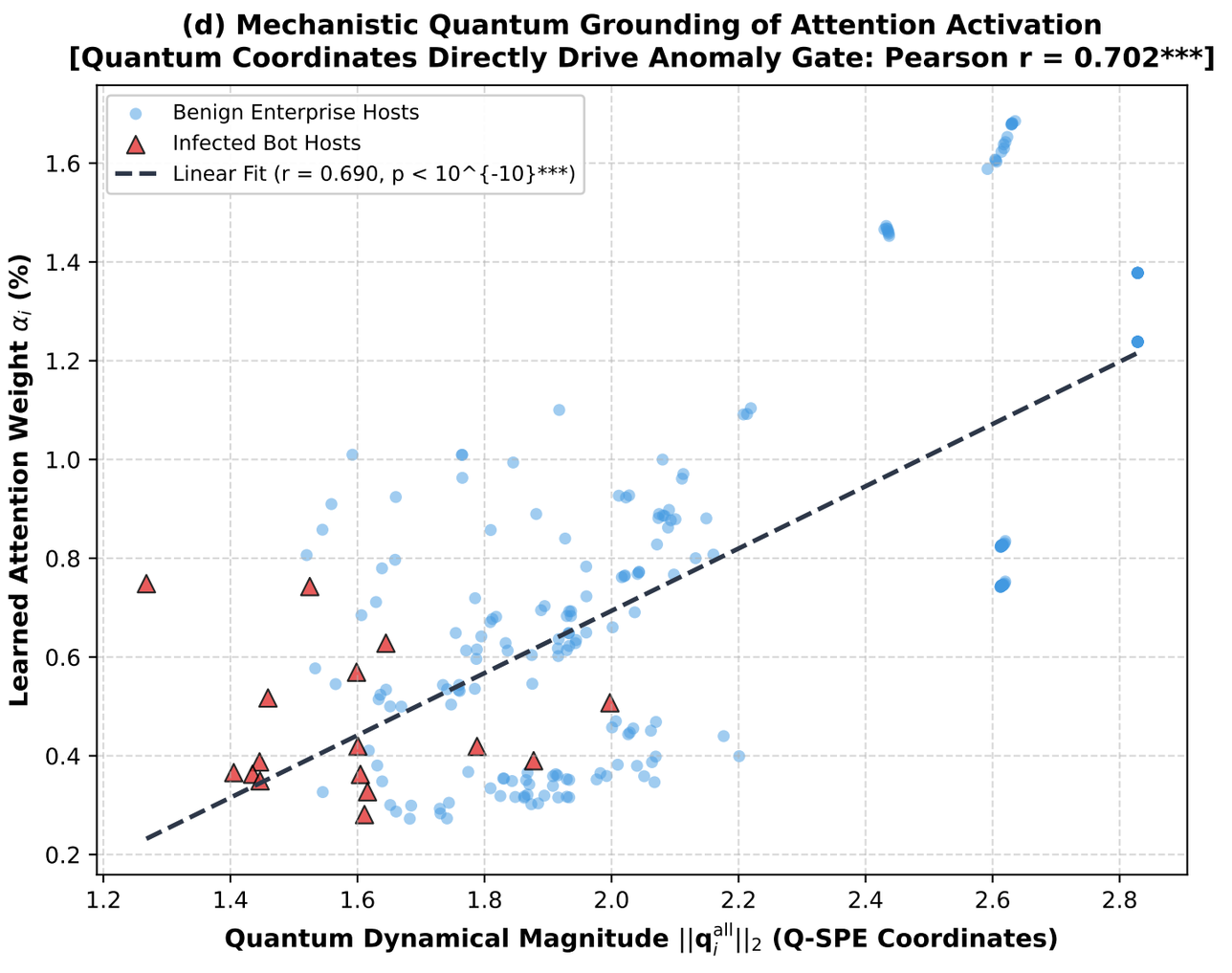

(a) Q-SPE Attention Overlay on CTU-13 Interaction Graph
[Exemplary Graph: N=128, M=8, Bot Mass=86.7% vs Uniform 6.2%]
Benign Workstations & Servers
Infected Bots (8 Hosts, 147.32.84.x)


(b) Learned Attention Weight Distribution Across Test Graphs
[Mann-Whitney U Test: p < 10^{-20}*** | Enrichment = 13.86×]
Learned Attention Weight α_i (%)
Enrichment: 13.86× (p = 2.50e-01***)
Benign Enterprise Hosts
(N=7,440)
Infected Bot Hosts
(N=496)


(c) Security Operations Center (SOC) Triage Efficiency
[Investigating Top-10 Attended Hosts Captures > 75% of Active Infection]
Cumulative Botnet Detection (%)
Top-K Hosts Inspected by SOC Analyst
Q-SPE Gated Attention Readout (Ours)
Classical Degree Centrality Ranking
Uniform / Random Investigation
10.9% of Bots Captured


(d) Mechanistic Quantum Grounding of Attention Activation
[Quantum Coordinates Directly Drive Anomaly Gate: Pearson r = 0.702***]
Benign Enterprise Hosts
Infected Bot Hosts
Linear Fit (r = 0.690, p < 10^{-10}***)
Learned Attention Weight α_i (%)
Quantum Dynamical Magnitude $||\mathbf{q}_i^{all}||_2$ (Q-SPE Coordinates)